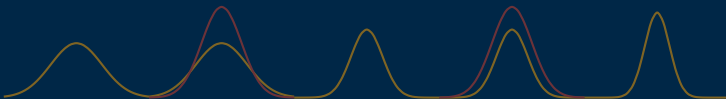


Gated Data-Space Inversion for Reservoir Prediction

Reducing variance loss in posteriors

Masters dissertation ▷ Thiago Tomás de Paula



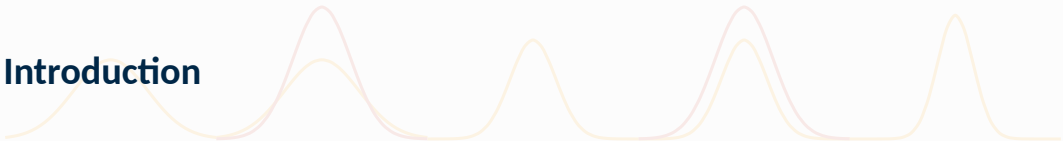
August 4th, 2026

Graduate Program in Mechanical Engineering

University of Brasília



Introduction



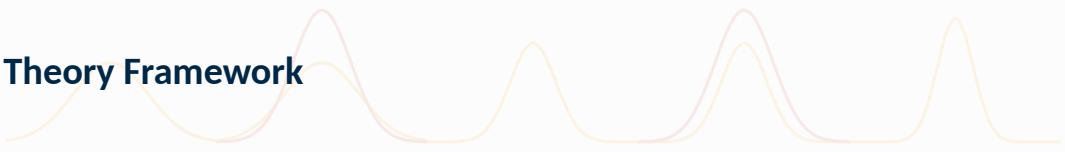








Theory Framework

The image features two decorative, wavy lines that span the width of the page. One line is a light yellow color, and the other is a light pink color. They are layered on top of each other, creating a soft, flowing background element. The lines have multiple peaks and valleys, giving them a rhythmic, wave-like appearance.







$$\mathbf{d}_j^{\ell+1} = \mathbf{d}_j^{\ell} + \mathbf{K}^{\ell}(\mathbf{d}_{\text{obs}} + \sqrt{\alpha^{\ell}} \mathbf{e}_j^{\ell} - \mathbf{d}_{h,j}^{\ell}), \quad (1a)$$

$$\mathbf{K}^{\ell} = \Delta \mathbf{D}^{\ell} \Delta \mathbf{D}_h^{\ell} (\Delta \mathbf{D}_h^{\ell} \Delta \mathbf{D}_h^{\ell} + \alpha^{\ell} \mathbf{C}_e)^{-1}. \quad (1b)$$



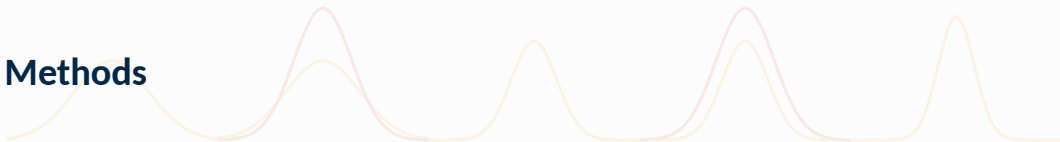
$$\mathbf{d}_j^{\ell+1} = \mathbf{d}_j^{\ell} + (\mathbf{R} \circ \mathbf{K}^{\ell})(\mathbf{d}_{\text{obs}} + \sqrt{\alpha^{\ell}} \mathbf{e}_j^{\ell} - \mathbf{d}_{h,j}^{\ell}), \quad (2)$$

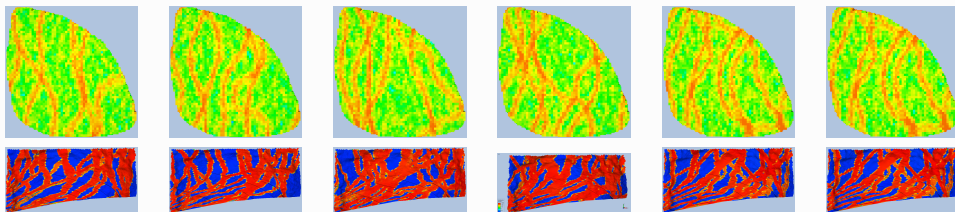


$$\mathbf{d}_j^{\ell+1} = \mathbf{d}_j^\ell + (\mathbf{R} \circ \mathbf{K}^\ell)[\boldsymbol{\alpha}_j^\ell \circ (\mathbf{d}_{\text{obs}} - \mathbf{d}_{h,j}^\ell)], \quad (3a)$$

$$\alpha_{j,v}^\ell = \text{erf} \left(\frac{1}{\lambda \sqrt{2\sigma_v^2}} |(\mathbf{d}_{\text{obs}} - \mathbf{d}_{h,j}^\ell)_v| \right) \quad (3b)$$

Methods

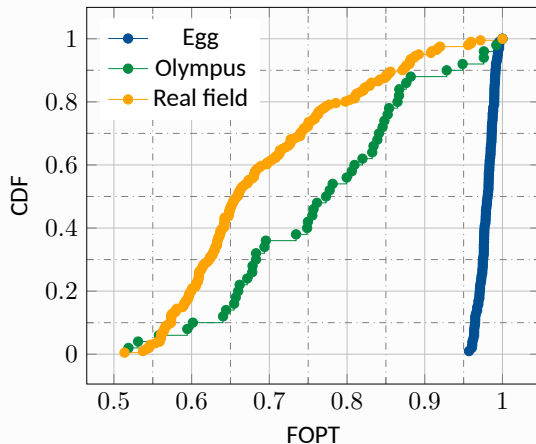


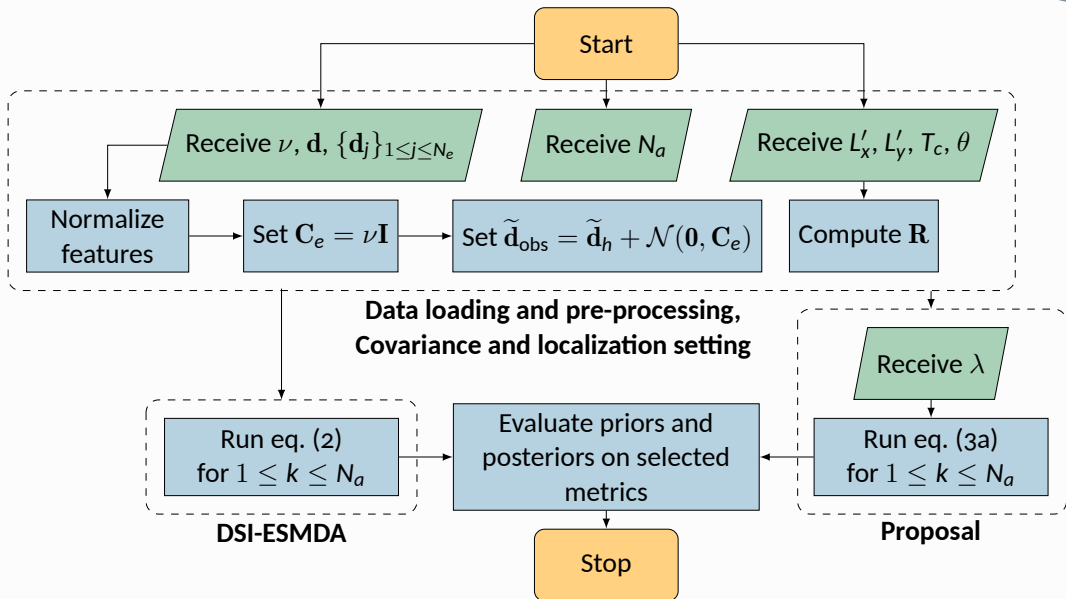


Property	Egg	Olympus	Real field	Unit
Rock compressibility	0.0	1.42×10^{-10}	-	1/Pa
Water compressibility	1.0×10^{-10}	3.97×10^{-10}	-	1/Pa
Oil dynamic viscosity	5.0×10^{-3}	$2.8-3.5 \times 10^{-3}$	-	Pa·s
Water dynamic viscosity	1.0×10^{-3}	0.398×10^{-3}	-	Pa·s
Well diameter	20	19	-	cm
Wells	8(P) 4(I)	11(P) 7(I)	16(P) 8(I)	-
Active cells	18,533	192,750	-	-
Cell size	$8 \times 8 \times 4$	$50 \times 50 \times 3$	-	m×m×m
Ensemble size	101	50	201	-



The field oil production total (FOPT) is a measurement commonly used to compare ensembles since ground-truth and realizations are each represented by a single value: how much oil was produced within the period of interest.







Four metrics are considered on the evaluation of posterior/prior distributions [1, subsection 9.4.2]: history observation coverage, validation observation coverage, mean squared error, and data-mismatch. Denoting an ensemble by $\mathbf{Y} = [\mathbf{y}_1 \ \cdots \ \mathbf{y}_{N_e}]$, these metrics are defined as:

$$\text{OC}_h(\mathbf{Y}) = \frac{1}{N_h} \sum_{i=1}^{N_h} \mathbb{1} (\min(\mathbf{Y})_i \leq \mathbf{d}_{h,i} \leq \max(\mathbf{Y})_i) \quad (4)$$

$$\text{OC}_v(\mathbf{Y}) = \frac{1}{N_v} \sum_{i=1}^{N_v} \mathbb{1} (\min(\mathbf{Y})_i \leq \mathbf{d}_{v,i} \leq \max(\mathbf{Y})_i) \quad (5)$$

$$\text{MSE}(\mathbf{Y}) = \frac{1}{N_h} (\mathbf{d}_h - \bar{\mathbf{y}})^\top \mathbf{C}^{-1} (\mathbf{d}_h - \bar{\mathbf{y}}), \quad \mathbf{C} = \Delta \mathbf{Y} \Delta \mathbf{Y}^\top + \mathbf{C}_e \quad (6)$$

$$\mathcal{O}(\mathbf{y}_j) = \frac{1}{2N_h} (\mathbf{d}_h - \mathbf{y}_{j,h})^\top \mathbf{C}_e^{-1} (\mathbf{d}_h - \mathbf{y}_{j,h}) \quad (7)$$

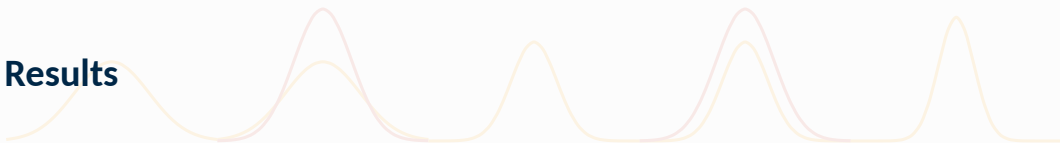


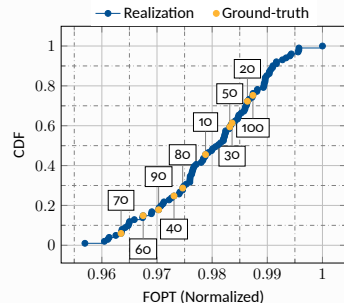
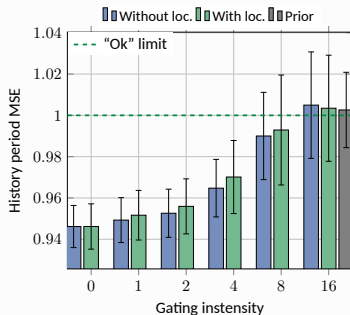
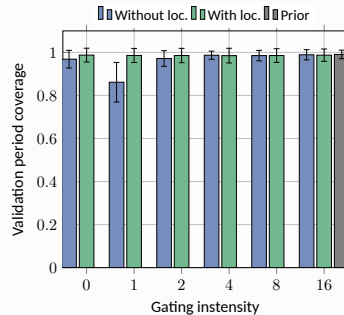
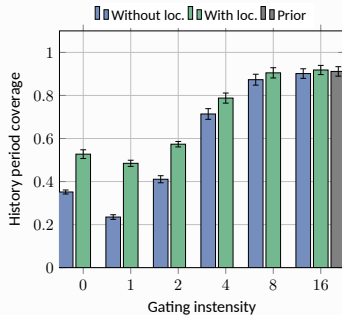
Reservoir	N_e	Number of time steps	Time step size	Simulation life	History split
Egg	100	120	30 days	3600 days (9.8 years)	60%
Olympus	49	20	1 calendar year	7305 days (20 years)	60%
Real field	200	510	1 calendar month	12785 days (35 years)	60%*

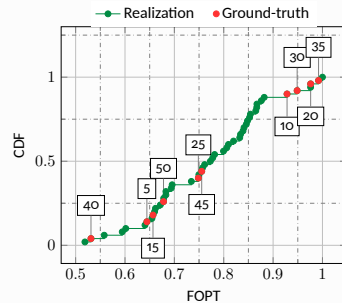
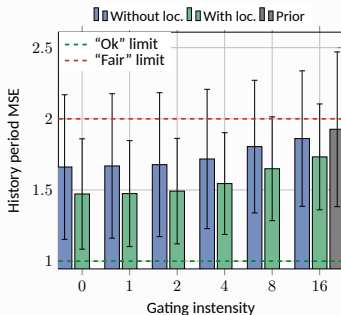
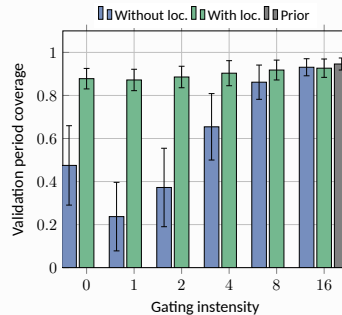
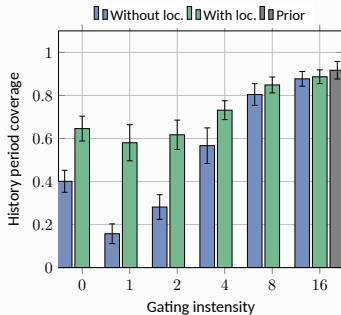
Reservoir	Critical length in x	Critical length in y	Critical time	Anisotropy
Egg	100	100	3000 days	0°
Olympus	100	100	7200 days	0°
Real field	-	-	10500 days	0°

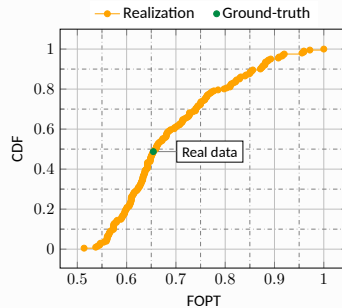
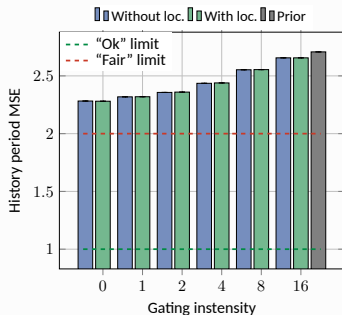
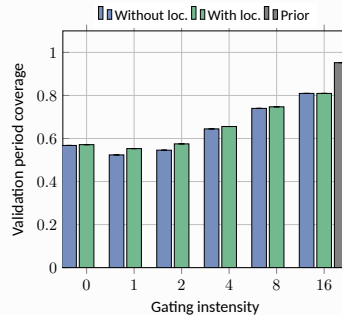
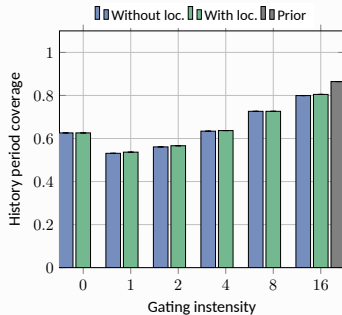
Reservoir	Well features used		Number of wells		Total features
	Producer	Injector	Producer	Injector	
Egg	OPR, WPR	BHP	4	8	16
Olympus	OPR, WPR	WIR	11	7	29
Real field	BHP, WCTP	BHP	16	8	39 (a WCTP was constant)

Results

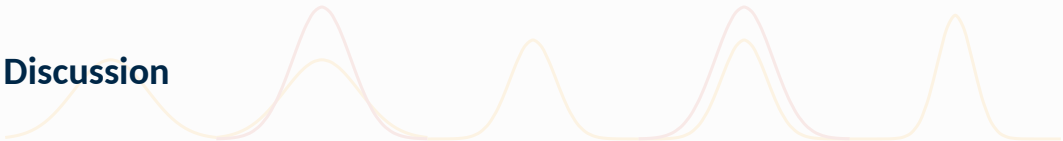








Discussion

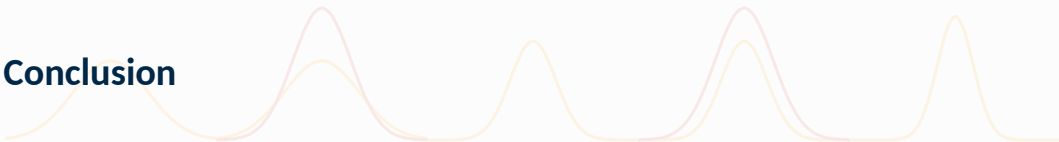








Conclusion













- [1] Alexandre Anozé Emerick. *Ensemble Data Assimilation Applied to Geological Reservoir Models*. 1st ed. Rio de Janeiro, RJ, Brazil: Universidade Petrobras, June 2025. ISBN: 978-65-88763-29-2.

Gated DSI-ESMDA

Reducing variance loss in posteriors

Thiago Tomás de Paula

Graduate students at University of Brasília, Brazil

August 4th, 2026

“It’s hard to make predictions, specially about the future”

— Yogi Berra

Thank you!